MA1522 Linear Algebra for Computing
L2 Lesson 20



Tan Kai Meng

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Question 1

PollEv.com/kaimengtan612

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation such that

$$T\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}2\\1\end{pmatrix}, \qquad T\begin{pmatrix}0\\1\\1\end{pmatrix} = \begin{pmatrix}1\\3\end{pmatrix}.$$

For each of the following vectors $\boldsymbol{v}$, determine whether $T(\boldsymbol{v})$ can be uniquely determined from the given information. If so, find $T(\boldsymbol{v})$.

a. $\boldsymbol{v} = \begin{pmatrix}2\\3\\1\end{pmatrix}$.

b. $\boldsymbol{v} = \begin{pmatrix}1\\1\\1\end{pmatrix}$.

c. $\boldsymbol{v} = \begin{pmatrix}x\\y\\z\end{pmatrix}$, where $y = x + z$.

Question 1(a,b): Solution

(a) Since $\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$,

$$T\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = T\left(2\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right) = 2T\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + T\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$
$$= 2\begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}.$$

(b) Suppose $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = a\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$. Then $a = 1$, $a + b = 1$ and $b = 1$, which is inconsistent.

Thus $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \notin \operatorname{span}\left\{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right\}$, and $T(\boldsymbol{v})$ cannot be determined from the given information.

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Question 1(c): Solution

- Since $y = x + z$, we have

$$\boldsymbol{v} = \begin{pmatrix} x \\ x+z \\ z \end{pmatrix} = x \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

Thus,

$$T(\boldsymbol{v}) = T\left(x \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right) = xT \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + zT \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

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Kernel and Nullity

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation.
The **kernel of** T, denoted $\ker(T)$, is

$$\ker(T) := \{\boldsymbol{u} \in \mathbb{R}^n \mid T(\boldsymbol{u}) = \mathbf{0}\}.$$

Theorem

1. $\ker(T) = \text{Null}([T])$.
2. $\ker(T)$ *is a subspace of* $\mathbb{R}^n$.

The **nullity of** T, denoted $\text{nullity}(T)$, is the dimension of $\ker(T)$, i.e.

$$\text{nullity}(T) = \dim(\ker(T)).$$

Corollary

$$\text{nullity}(T) = \text{nullity}([T]).$$

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Range and Rank

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation.
The **range of** T, denoted $\mathrm{R}(T)$, is

$$\mathrm{R}(T) := \{\boldsymbol{v} \in \mathbb{R}^m \mid \boldsymbol{v} = T(\boldsymbol{u}) \text{ for some } \boldsymbol{u} \in \mathbb{R}^n\} = \{T(\boldsymbol{u}) \mid \boldsymbol{u} \in \mathbb{R}^n\}.$$

Theorem

1. $\mathrm{R}(T) = \mathrm{Col}([T])$.
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The **rank of** T, denoted $\mathrm{rank}(T)$, is the dimension of $\mathrm{R}(T)$, i.e.

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Rank-Nullity Theorem

Theorem (Rank-Nullity Theorem)

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then

$$\operatorname{rank}(T) + \operatorname{nullity}(T) = n.$$

Question 2

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For each of the following linear transformations, find its kernel and range, write down its rank and nullity, and verify the rank-nullity theorem.

a. $T_1 : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ y-x \end{pmatrix}$.

b. $T_2 : \mathbb{R}^2 \to \mathbb{R}^3$ such that $T_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 0 \\ 0 \end{pmatrix}$.

c. $T_3 : \mathbb{R}^5 \to \mathbb{R}$ such that $T_3 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = x_3 + 2x_4 - x_5$.

Question 2(a,b): Solution

a. $[T_1] = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$; $\det([T_1]) = 2$ so $[T_1]$ is invertible.
Thus $\ker(T_1) = \text{Null}([T_1]) = \{\mathbf{0}\}$ and $\text{R}(T_1) = \text{Col}([T_1]) = \mathbb{R}^2$.
So $\text{nullity}(T_1) = 0$ and $\text{rank}(T_1) = 2$, and
$\text{rank}(T_1) + \text{nullity}(T_1) = 2 + 0 = 2$.

b. $[T_2] = \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$.

$$\ker(T_2) = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \mid x + y = 0 \right\} = \left\{ \begin{pmatrix} x \\ -x \end{pmatrix} \mid x \in \mathbb{R} \right\} = \text{span}\left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}.$$

$$\text{R}(T_2) = \text{Col}([T_2]) = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}.$$

So $\text{Null}(T_2) = 1$, $\text{rank}(T_2) = 1$, and
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So $\text{Null}(T_2) = 1$, $\text{rank}(T_2) = 1$, and
$\text{rank}(T_2) + \text{nullity}(T_2) = 1 + 1 = 2$.

Question 2(c): Solution

(c) $$\ker(T_3) = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} \middle| \, x_3 + 2x_4 - x_5 = 0 \right\}$$

$$= \left\{ \begin{pmatrix} x_1 \\ x_2 \\ -2x_4+x_5 \\ x_4 \\ x_5 \end{pmatrix} \middle| \, x_1, x_2, x_4, x_5 \in \mathbb{R} \right\}$$

$$= \operatorname{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

$\mathrm{R}(T_3) = \mathbb{R}$.

Thus $\operatorname{nullity}(T_3) = 4$ and $\operatorname{rank}(T_3) = 1$, and $\operatorname{rank}(T_3) + \operatorname{nullity}(T_3) = 1 + 4 = 5$.

One-to-one and Onto

Theorem

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. The following statements are equivalent:

1. *T is one-to-one.*
2. $\ker(T) = \{\mathbf{0}_n\}$.
3. $\text{nullity}(T) = 0$.

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Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. The following statements are equivalent:

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Question 3

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For each of the following linear transformations, determine whether it is one-to-one and/or onto.

a. $T : \mathbb{R}^4 \to \mathbb{R}^6$ such that $\text{rank}(T) = 4$.

b. $T : \mathbb{R}^6 \to \mathbb{R}^4$ such that $\text{nullity}(T) = 2$.

c. $T : \mathbb{R}^4 \to \mathbb{R}^6$ such that the reduced row-echelon form of its standard matrix has 3 nonzero rows.

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a. $T : \mathbb{R}^4 \to \mathbb{R}^6$ such that $\text{rank}(T) = 4$.

   $\text{rank}(T) = 4 < 6$, T is not onto.
   $\text{nullity}(T) = 4 - \text{rank}(T) = 0$, so T is one-to-one.

b. $T : \mathbb{R}^6 \to \mathbb{R}^4$ such that $\text{nullity}(T) = 2$.

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b. $T : \mathbb{R}^6 \to \mathbb{R}^4$ such that $\text{nullity}(T) = 2$.

   $\text{nullity}(T) = 2 > 0$, so T is not one-to-one.
   $\text{rank}(T) = 6 - \text{nullity}(T) = 4$, so T is onto.

c. $T : \mathbb{R}^4 \to \mathbb{R}^6$ such that the reduced row-echelon form of its standard matrix has 3 nonzero rows.

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c. $T : \mathbb{R}^4 \to \mathbb{R}^6$ such that the reduced row-echelon form of its standard matrix has 3 nonzero rows.

   $\text{rank}(T) = \text{rank}([T]) = \text{rank}(\text{RREF}([T])) = 3 < 6$, so T is not onto.
   $\text{nullity}(T) = 4 - \text{rank}(T) = 1 > 0$, so T is not one-to-one.

Next Lesson ...

We will revise **Chapters 1 and 2**.
There will also be a **briefing on your final examination**.